

# Food Science Labs





## About the Course

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Food Science



## Scheduling

| GRADE | SCHEDULE INFO.                          | BOOKS |
|-------|-----------------------------------------|-------|
| 9-12  | Food Science Labs<br>1 time/week 60 min |       |

### Sample Weekly View

| Day 1                                                           | Day 2                | Day 3                | Day 4                                                   | Day 5                                     |
|-----------------------------------------------------------------|----------------------|----------------------|---------------------------------------------------------|-------------------------------------------|
| <b>Science: Food Science</b>                                    |                      |                      |                                                         |                                           |
| Food Science Lessons<br>Nature Walks &<br>Scouting: Grades 9-12 | Food Science Lessons | Food Science Lessons | Food Science Lessons<br>Nature Notebook:<br>Grades 9-12 | Food Science Lessons<br>Food Science Labs |



## Planning & Prep

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**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

[Click THIS text  
or scan the QR  
code for links.](#)

QR

# Food Science Labs

## How To Approach



### Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



### Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



### Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

# Food Science Labs

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## Term 1

### WEEK 1

#### 60m Food Science Labs - Lesson 1

Materials: Salt, Fat, Acid, Heat, ingredients for Caesar Salad p.48, protein, bread, indicated kitchen supplies

##### → KITCHEN LAB

We'll learn more about salads in the coming weeks, but for this week, make Caesar Salad as on p.48-49 in SFAH. You may either use store-bought mayo for now or try the recipe (visual p.86-87 or traditional p.375). Serve alongside your choice of protein and bread/croutons. If you want to make your own mayonnaise, you might find the video below helpful.

∞ Video Link: Mayonnaise and the Science of Emulsions

### WEEK 2

#### 60m Food Science Labs - Lesson 2

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Salt Lesson. Decide on next Salt Lesson p.428 and add to shopping list, if you haven't already.

### WEEK 3

#### 60m Food Science Labs - Lesson 3

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Salt Lesson. Decide on next Salt Lesson p.428 and add to shopping list, if you haven't already.

### WEEK 4

#### 60m Food Science Labs - Lesson 4

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Salt Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

### WEEK 5

#### 60m Food Science Labs - Lesson 5

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

### WEEK 6

#### 60m Food Science Labs - Lesson 6

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

### WEEK 7

#### 60m Food Science Labs - Lesson 7

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

### WEEK 8

#### 60m Food Science Labs - Lesson 8

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

### WEEK 9

#### 60m Food Science Labs - Lesson 9

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

##### → KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

| Term 1                                                                                                                                                                                                                                                                                                                                                                                |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>WEEK 10</div><div><div><div></div></div><div>60m Food Science Labs - Lesson 10</div></div><div><div></div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div><div><div>→ KITCHEN LAB</div><div>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</div></div></div> |  |
| <div><div>WEEK 11</div><div><div><div></div></div><div>60m Food Science Labs - Lesson 11</div></div><div><div></div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div><div><div>→ KITCHEN LAB</div><div>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</div></div></div> |  |
| <div><div>WEEK 12</div><div><div><div></div></div><div>60m Food Science Labs - Lesson 12</div></div><div><div></div>Exam: Dinner Night</div><div><div>→ EXAMS</div><div>Answer exam questions from:<br/>Kitchen Lab experience</div></div></div>                                                                                                                                      |  |

# Food Science Labs

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## Term 2

|               |                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEEK 1</b> | <b>60m Food Science Labs - Lesson 13</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |
| <b>WEEK 2</b> | <b>60m Food Science Labs - Lesson 14</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |
| <b>WEEK 3</b> | <b>60m Food Science Labs - Lesson 15</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |
| <b>WEEK 4</b> | <b>60m Food Science Labs - Lesson 16</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>Prep: Read Teacher Tip</p> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> | <p>★ <b>TEACHER TIP</b><br/>Students are conducting citizen science next week and will not have their regular dinner night. They may need to adjust their schedule, so that their Kitchen Waste Audit is completed at the end of the week.</p> |
| <b>WEEK 5</b> | <b>60m Food Science Labs - Lesson 17</b><br><div>Materials: internet access, kitchen scale</div> <p>→ <b>DATA COLLECTION</b><br/>Conduct the second Kitchen Waste Audit. Adjust your schedule, if needed, so that this audit is completed after all 4 Daily Food Waste Logs.<br/>∞ Website Link: Food Conservation</p>        |                                                                                                                                                                                                                                                |
| <b>WEEK 6</b> | <b>60m Food Science Labs - Lesson 18</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |
| <b>WEEK 7</b> | <b>60m Food Science Labs - Lesson 19</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |
| <b>WEEK 8</b> | <b>60m Food Science Labs - Lesson 20</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |
| <b>WEEK 9</b> | <b>60m Food Science Labs - Lesson 21</b><br><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <p>→ <b>KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p>                               |                                                                                                                                                                                                                                                |



| Term 2  |                                                                                                                                                                                                                                                                                                                                              |  |
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| WEEK 10 | <div><div><div></div></div><div><b>60m Food Science Labs - Lesson 22</b></div></div> <div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <div><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</div> |  |
| WEEK 11 | <div><div><div></div></div><div><b>60m Food Science Labs - Lesson 23</b></div></div> <div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div> <div><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</div> |  |
| WEEK 12 | <div><div><div></div></div><div><b>60m Food Science Labs - Lesson 24</b></div></div> <div>Exam: Dinner Night</div> <div><b>→ EXAMS</b><br/>Answer exam questions from:<br/>Kitchen Lab experience</div>                                                                                                                                      |  |



## Term 3

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| <b>WEEK 1</b> | <p><b>60m Food Science Labs - Lesson 25</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 2</b> | <p><b>60m Food Science Labs - Lesson 26</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 3</b> | <p><b>60m Food Science Labs - Lesson 27</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 4</b> | <p><b>60m Food Science Labs - Lesson 28</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 5</b> | <p><b>60m Food Science Labs - Lesson 29</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 6</b> | <p><b>60m Food Science Labs - Lesson 30</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 7</b> | <p><b>60m Food Science Labs - Lesson 31</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 8</b> | <p><b>60m Food Science Labs - Lesson 32</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
| <b>WEEK 9</b> | <p><b>60m Food Science Labs - Lesson 33</b></p> <p>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</p> <p><b>→ KITCHEN LAB</b><br/>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</p> |  |
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| Term 3  |                                                                                                                                                                                                                                                                                                                                                                          |  |
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| WEEK 10 | <div><div><div></div></div><div><b>60m Food Science Labs - Lesson 34</b></div></div> <div><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div></div> <div><div>→ <b>KITCHEN LAB</b></div><div>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</div></div> |  |
| WEEK 11 | <div><div><div></div></div><div><b>60m Food Science Labs - Lesson 35</b></div></div> <div><div>Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies</div></div> <div><div>→ <b>KITCHEN LAB</b></div><div>Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.</div></div> |  |
| WEEK 12 | <div><div><div></div></div><div><b>60m Food Science Labs - Lesson 36</b></div></div> <div><div>Exam: Dinner Night</div></div> <div><div>→ <b>EXAMS</b></div><div>Answer exam questions from:<br/>Kitchen Lab experience</div></div>                                                                                                                                      |  |